

Session 6 — Solutions and facilitation notes

Foundations of Clinical Research · instructor copy (do not distribute before the session)

Exercise numbers follow `Practicals/s6_exercises.md`. All FLUID-ICU numbers come from `Course/Data/key_results.md` (simulated data).

Exercise 1 — True or false

#	Answer	One-line explanation
1	T	That is the practical reading of a 95% CI: the range of values compatible with the data.
2	F	Absence of evidence is not evidence of absence. The CI (−15.2 to 41.6 ml/kg) is wide: the pilot cannot tell.
3	F	p is calculated <i>assuming</i> the null is true; it is the probability of data at least this extreme under the null, not the probability of the null.
4	T	A 95% CI that includes the null value corresponds to $p > 0.05$ (two-sided), and vice versa.
5	T	More patients → more precision → narrower CI.
6	F	A tiny p can accompany a trivial effect in a large study. Judge size from the estimate and CI.
7	T	This is the strict definition: the 95% refers to the method over repeated studies (the "dance of the CIs").
8	T	A small study has low power; a real effect can be missed (missed fire).
9	T	Alpha = type I error rate = risk of a false alarm.
10	F	p-values do not "trend". Report the estimate and CI and call it what it is: not statistically significant.
11	T	With 20 independent tests and no real effect, the chance of at least one $p < 0.05$ is about 64% (the jelly-bean problem).
12	F	Same patients measured twice: use a paired test (paired t-test, or Wilcoxon signed-rank if skewed).

Exercise 2 — Choose the test

#	Test	Why
a	Independent (unpaired) t-test	Numerical, roughly symmetric outcome, two independent groups.
b	Mann–Whitney U	Numerical but skewed, small groups. (Same choice as Paper 1 for fluid balance.)
c	Chi-square test	Binary outcome, two independent groups, large expected counts.
d	Fisher's exact test	Binary outcome, very small counts (expected counts < 5).
e	Paired t-test	Same patients measured twice; differences roughly symmetric. (Wilcoxon signed-rank if skewed.)
f	McNemar test	Binary outcome measured twice in the same patients (paired categorical).

Facilitation: if someone suggests "chi-square" for (f), ask "are these two independent groups?" → no, same patients → paired → McNemar.

Exercise 3 — Match r to the scatter plots

Plot	r	Why
A	-0.58	Points fall from left to right, moderate scatter
B	0.01	Round cloud, no straight-line trend
C	0.86	Tight rising band
D	0.30	Weak rising trend, lots of scatter

(The plotted data were simulated with exactly these correlations.)

Bonus: **no**, not on correlation alone. High r means the two methods rise and fall together, not that they give the same value. Look at **agreement**: a Bland–Altman plot (difference against average), the mean difference (bias) and the 95% limits of agreement, and decide whether those differences are clinically acceptable.

Exercise 4 — [L2] Interpret your group's comparison

Numbers from `Course/Data/key_results.md` (all unadjusted; simulated data).

Q	Test	Result	Model sentence
Q1	Chi-square (binary outcome, two large groups)	134/373 (35.9%) vs 45/245 (18.4%); RD 17.6 points (95% CI 10.7 to 24.4); RR 1.96; p < 0.001	"Twenty-eight-day mortality was 35.9% with a positive and 18.4% with a negative balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4)."

Q	Test	Result	Model sentence
Q2	Chi-square	AKI 230/373 (61.7%) vs 71/245 (29.0%); $p < 0.001$; gradient 29%, 54%, 63%, 75% across balance groups	"AKI occurred in 61.7% of patients with a positive and 29.0% with a negative balance."
Q3	Mann–Whitney (count, piled at 0 and 28)	Median 20 (0 to 28) vs 28 (22 to 28) days; $p < 0.001$	"Patients with a positive balance had fewer ventilator-free days (median 20 vs 28)."
Q4	Mann–Whitney (lactate skewed)	Median lactate 3.1 (2.2 to 4.5) in non-survivors vs 2.3 (1.6 to 3.4) mmol/L in survivors; mortality < 2 : 19%, 2–4: 31%, > 4 : 50%	"Non-survivors had higher admission lactate (median 3.1 vs 2.3 mmol/L)."
Q5	Mann–Whitney (skewed, survivors)	Median 7.5 (5.2 to 11.2) vs 4.8 (3.3 to 7.1) days; Hodges–Lehmann difference 2.5 days (95% CI 1.8 to 3.2); $p < 0.001$	"Among survivors, ICU stay was longer with a positive balance (median 7.5 vs 4.8 days; difference 2.5 days, 95% CI 1.8 to 3.2)." Note: the t-test difference in means (2.7 days) is the wrong summary for this skewed outcome.

Reason for bias (all groups): confounding by indication — the positive-balance group was sicker at baseline (Table 1).

Exercise 5 — Read a regression table

The table is mock data; p-values are consistent with the CIs (computed from the log-scale standard error).

1. **Binary outcome** (delirium yes/no) → logistic regression, which reports odds ratios.
2. **1** (a ratio). An OR of 1 means equal odds.
3. CI excludes 1 (☐) or includes 1 (☐):

Row	OR (95% CI)	Excludes 1?
Benzodiazepine	2.10 (1.40–3.20)	☐
Age per 10 y	1.30 (1.12–1.51)	☐
Mechanical ventilation	1.60 (0.95–2.70)	☐
Female sex	0.85 (0.60–1.20)	☐
Dementia	3.40 (1.90–6.10)	☐
Sleep protocol	0.62 (0.41–0.94)	☐

4. Model sentence: "After adjusting for age, sex, ventilation, dementia, sleep protocol and illness severity, patients who received a benzodiazepine infusion had about twice the odds of developing delirium (OR 2.10, 95% CI 1.40 to 3.20)."

5. The OR multiplies for each 10 years: 20 years older → $1.30 \times 1.30 \approx 1.69$ times the odds (all else equal).

6. **Mechanical ventilation** and **female sex** (CIs include 1).

7. Problems in the conclusion:

- **Causal language** ("cause", "reduces") from an observational cohort: should be "associated with". Confounding by indication is likely (sicker or more agitated patients receive benzodiazepines).
- **"Trend"** for $p = 0.08$: the CI (0.95–2.70) includes 1; it is not statistically significant. Report the estimate and CI.
- **"Reduces delirium by 38%"** confuses odds with risk (OR 0.62 is a 38% lower *odds*, not risk; delirium is common, so the risk reduction is smaller) and again implies causation. The sleep protocol was not randomised.

Exercise 6 — Read a Kaplan–Meier curve (simulated trial; values computed from the plotted data)

1. About **82%** alive in the restrictive group and about **67%** in the usual-care group at day 28 (read where each curve ends). Accept $\pm 3\%$.
2. Each tick is a **censored** patient: follow-up stopped (lost to follow-up, transferred, withdrew) without death being observed. They count as alive up to that day.
3. **67** usual-care patients were at risk at day 21. The number falls because of **deaths and censoring**; in the restrictive group 16 died but 33 left the risk set ($100 \rightarrow 67$ at day 28) because 17 were censored before day 28.
4. The log-rank test gives evidence that the two survival curves differ ($p = 0.009$: results this different would be unusual if the strategies had the same survival). It does **not** give the size of the effect, its 95% CI, or any adjustment for confounders (and in a trial, it is not "proof" of causation without a good design).
5. **Cox regression** gives the hazard ratio with a 95% CI. Check the **proportional-hazards** assumption: curves that cross or converge strongly suggest the HR is not constant over time.
6. No: without numbers at risk the reader cannot see how many patients support each part of the curve; the right-hand end may rest on very few patients. Reporting guidelines and reviewers expect them.

Exercise 7 — Why is it missing? MCAR, MAR or MNAR

#	Mechanism	Consequence for complete-case analysis
a	MCAR — an accident unrelated to the patients	Unbiased, just less precise (fewer patients)
b	MAR — depends on admission time, which is recorded	Possibly biased if night admissions differ; multiple imputation using admission time can correct it
c	MNAR — depends on the missing value itself (the worst pain)	Biased (pain looks better than it is); no guaranteed fix: sensitivity analysis

#	Mechanism	Consequence for complete-case analysis
d	MNAR (deteriorating patients are transferred); at best MAR if the reason for transfer is recorded	Biased (mortality underestimated); chase day-28 status by phone/records; sensitivity analysis
e	MCAR (missing by design for one month, unrelated to patients) — unless patients in the first month differed	Usually unbiased; check whether early patients differed
f	MAR — depends on age, recorded for all	Biased towards younger patients' quality of life; multiple imputation including age

Bonus (d): report the main result plus a **best-case** (all transferred patients alive) and **worst-case** (all died) analysis, or vary the assumed mortality among transferred patients; if the conclusion holds under all scenarios, it is robust.

Exercise 8 — [L2] Interpret your group's model

Numbers from `Course/Data/key_results.md` (simulated data).

Q	Crude	Adjusted (complete case, n = 527 unless stated)	Missing-data sensitivity
Q1	OR 2.49 (1.70 to 3.70), n = 618; same 527 patients 2.22 (1.48 to 3.39)	OR 1.62 (1.02 to 2.59), p = 0.044; Cox HR crude 2.17 (1.55 to 3.05) → adjusted 1.46 (1.00 to 2.13), p = 0.053; PH test p = 0.10	MI (m = 20, n = 620): OR 1.70 (1.09 to 2.64), p = 0.020
Q2	OR per 10 ml/kg 1.37 (1.27 to 1.47)	1.27 (1.16 to 1.40), p < 0.001	—
Q3	−1.48 (−1.84 to −1.13) days per 10 ml/kg	−0.61 (−1.00 to −0.21) days, p = 0.003	—
Q4	OR per 1 mmol/L: CC 1.38 (1.23 to 1.55) vs MI 1.44 (1.28 to 1.61)	Adjusted (age, sex, diabetes, CKD, HF): CC 1.39 (1.24 to 1.57), n = 527 vs MI 1.45 (1.29 to 1.64), n = 620; + SOFA: CC 1.21 (1.05 to 1.39) vs MI 1.23 (1.07 to 1.42)	Lactate missing 93/620 (15.0%), mostly less sick patients (SOFA 5 vs 8; mortality 19.4% vs 30.6%); not MCAR, MAR plausible
Q5	Ratio of geometric means 1.50 (1.34 to 1.68)	1.24 (1.10 to 1.39), p < 0.001 (n = 366 survivors with complete data)	—

1. Q1 confounders (pre-specified in the SAP): age, SOFA, lactate, vasopressor, mechanical ventilation, CKD. AKI and dialysis were deliberately NOT adjusted for (they occur after day 3 — outcomes or mediators).
2. In every question the adjusted estimate is closer to "no effect" than the crude one: part of the crude association was severity (confounding by indication).
3. All adjusted CIs above exclude the null, but Q1's lower limit is close to 1 (1.02; Cox 1.00): a modest, imprecise association.

4. Complete case 527 of 620; the left-out patients were mostly those without lactate, who were less sick.
5. MI changed the estimates only slightly (Q1 1.62 → 1.70; Q4 1.39 → 1.45): the conclusions are robust to how missing lactate was handled.
6. (Q1) Day-28 survival 81.6% (negative) vs 64.1% (positive); log-rank $p < 0.001$ (unadjusted). Adjusted OR 1.62 ($p = 0.044$) and HR 1.46 ($p = 0.053$) are in the same direction and of similar size with overlapping CIs; whether p falls just below or above 0.05 does not change the interpretation.

Model sentences (Q1): "Positive fluid balance was associated with higher 28-day mortality (crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment for age, SOFA, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, the association was smaller (adjusted OR 1.62, 95% CI 1.02 to 2.59)."

Exercise 9 — Proposal builder 6 (what a good answer looks like)

Sample size (running example): 40% vs 30% mortality, alpha 5% two-sided, power 80% → 356 per group (spreadsheet sheet 2); $/(1 - 0.10) \rightarrow 396$ per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report RR and risk difference with 95% CIs. Observational designs: adjusted OR (logistic) or HR (Cox) with confounders chosen in advance, about 1 model term per 10 events.

Missing-data plan (good enough): names the variables likely to be missing and why; prevention; main handling (complete case if little missing and plausibly MCAR, otherwise multiple imputation); one sensitivity analysis (e.g. complete case vs MI, or best/worst case for a missing outcome); reporting (n missing per variable, flow diagram). FLUID-ICU's SAP section 10 is a model.

Element	Good enough	Needs work
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test/model + effect measure with 95% CI	"Data will be analysed with SPSS"
Confounders	Named, justified, within events-per-variable	Stepwise selection; mediators adjusted
Missing data	Expected gaps + prevention + handling + sensitivity	Nothing, or "missing data will be excluded"